

Evaluating Central vs. Local Differential Privacy for Vector Embeddings in Medical RAG Systems

Problem Statement:

Retrieval augmented generation (RAG) reduces LLM hallucinations, but populating vector databases with sensitive corpora creates severe privacy vulnerabilities. Standard vector databases (e.g., Pinecone, Chroma) are optimized strictly for approximate nearest neighbor search and offer no inherent privacy protection. In a realistic threat scenario, such as an insider threat or a compromised database—an adversary gains white box access to raw embeddings. Our goal is to implement and evaluate a system that applies formal differential privacy to document embeddings, preventing an adversary from reconstructing source texts or inferring document inclusion, while quantifying the resulting degradation in QA utility. Our DP guarantee applies to the embedding layer; we treat LLM output privacy as out of scope for this work.

Approach:

Dataset & Models: We will utilize the open-source ChatDoctor dataset (synthetic medical QA conversations with sensitive health content) as our sensitive knowledge base. To maintain strict control over the embedding space geometry, we will use a local SentenceTransformer model (`all-MiniLM-L6-v2`) for vectorization.

Algorithms: We will implement and compare two formal privacy paradigms:

1. **Central DP:** Since embeddings are L2-normalized to the unit hypersphere, the global L2 sensitivity is bounded by 2. This enables calibrated Central DP via the Gaussian mechanism, adding noise to the database scaled as $\sigma = \frac{2\sqrt{2\ln(1.25/\delta)}}{\epsilon}$.
2. **Local DP (LDP):** We will compare this against a continuous Local Differential Privacy approach. We apply the Gaussian mechanism locally per document, with per-document sensitivity bounded by 2 under L2 normalization, consistent with the local model of DP (Kasiviswanathan & Smith, 2014), shifting the trust boundary away from the central server.

Evaluation:

Threat Model & Privacy Metrics: We assume a semi-honest adversary with white-box access to the vector database. We assume the adversary has knowledge of the embedding model, consistent with the white-box threat scenario. To evaluate privacy leakage, we will construct a Membership Inference Attack (MIA) using a 70/30 member/non-member split of ChatDoctor documents, utilizing shadow embeddings generated from the same SentenceTransformer to simulate the adversary’s auxiliary knowledge. Leakage will be quantified via the MIA’s Area Under the Curve (AUC) and the ROUGE similarity score of an embedding inversion attack. Embedding inversion will be approximated via nearest-neighbor text lookup against a held-out reference corpus, with ROUGE-L measuring reconstruction fidelity.

Utility Metrics: Utility will be measured by the end-to-end generation performance of the RAG system on held-out QA pairs. Because exact match accuracy is brittle for conversational medical QA, BERTScore will serve as our primary utility metric, supplemented by ROUGE-L. We will plot an empirical privacy-utility tradeoff curve for both the Central and Local DP mechanisms to compare their degradation profiles.